1 · Setup & Data Loading
1.1 Parse Resolutions & Token Mapping
1.2 Derive Categories from Tags
2 · Calibration Analysis (Snapshot Prices)
2.1 Overall Calibration Curve
2.2 Calibration Error by Snapshot Horizon
2.3 Favorite–Longshot Bias
2.4 Initial Price Calibration (Conditional)
3 · Category-Level Mispricings
3.1 Category Calibration Curves
3.2 Category × Price Zone Bias Heatmap
3.3 Aggregate Category Bias & Brier Score
3.4 Politics: Election vs Off-Cycle Markets
4 · Volume & Liquidity Analysis
4.1 Calibration by Volume Tier
4.2 Liquidity-Filtered Edge Survival
5 · Market Duration Effects
6 · Late-Stage Price Signals
7 · Temporal Stability of Edges
7.1 Rolling Monthly Bias for Top Segments
7.2 Edge Persistence Score
7.3 Is the Market Getting More Efficient?
8 · Exploitable Pattern Summary
8.1 Gross Edge Table
8.2 Transaction Cost–Adjusted Edge Table
8.3 Bootstrap Confidence Intervals
8.4 Kelly Criterion Position Sizing
8.5 Composite Exploitability Score
9 · Simulated Backtest
9.1 Cumulative P&L
9.2 Drawdown & Risk Metrics
9.3 P&L by Segment
10 · Bot Implementation Notes
Entry Signals
Position Sizing
Risk Limits
Key Findings

# Polymarket Bias & Mispricing Analysis

Code

# Closed Markets 2025 — Calibration, Exploitable Edges & Bot Strategy (v4)

Laurence

2026-02-19

# 1 · Setup & Data Loading

Show

Show

Show

```
## 111178 market rows loaded
```

Show

```
## 87 columns
```

## 1.1 Parse Resolutions & Token Mapping

Show

```
## 49753 binary Yes/No markets with clean resolution
```

Show

```
## Resolution rate (Yes = 1): 0.267
```

## 1.2 Derive Categories from Tags

Show

```
##
## ── Category distribution ──
```

Show

```
## # A tibble: 8 × 2
##   category          n
##   <chr>         <int>
## 1 Sports        14881
## 2 Crypto        14161
## 3 Politics       9844
## 4 Science/Tech   5570
## 5 Other          2410
## 6 Entertainment  2091
## 7 Economics       777
## 8 World Events     19
```

Show

```
##
## Politics: election-tagged = 4628 / 9844
```

# 2 · Calibration Analysis (Snapshot Prices)

The core question: **when the market prices a Yes token at probability $p$, does the event actually happen $p$% of the time?**

Show

```
## 49753 unique Yes tokens to look up
```

Show

```
## 100 batches of up to 500 tokens
```

Show

```
##   Batch 5/100 done (11595 snapshot rows so far)
##   Batch 10/100 done (23809 snapshot rows so far)
##   Batch 15/100 done (36470 snapshot rows so far)
##   Batch 20/100 done (48962 snapshot rows so far)
##   Batch 25/100 done (61591 snapshot rows so far)
##   Batch 30/100 done (74504 snapshot rows so far)
##   Batch 35/100 done (87249 snapshot rows so far)
##   Batch 40/100 done (100113 snapshot rows so far)
##   Batch 45/100 done (112551 snapshot rows so far)
##   Batch 50/100 done (123420 snapshot rows so far)
##   Batch 55/100 done (133622 snapshot rows so far)
##   Batch 60/100 done (145422 snapshot rows so far)
##   Batch 65/100 done (157735 snapshot rows so far)
##   Batch 70/100 done (169985 snapshot rows so far)
##   Batch 75/100 done (181988 snapshot rows so far)
##   Batch 80/100 done (194828 snapshot rows so far)
##   Batch 85/100 done (207703 snapshot rows so far)
##   Batch 90/100 done (220378 snapshot rows so far)
##   Batch 95/100 done (232904 snapshot rows so far)
##   Batch 100/100 done (245827 snapshot rows so far)
```

Show

```
##            used  (Mb) gc trigger  (Mb)  max used  (Mb)
## Ncells  3526771 188.4    5795942 309.6   5795942 309.6
## Vcells 43051343 328.5  121389701 926.2 121388229 926.2
```

Show

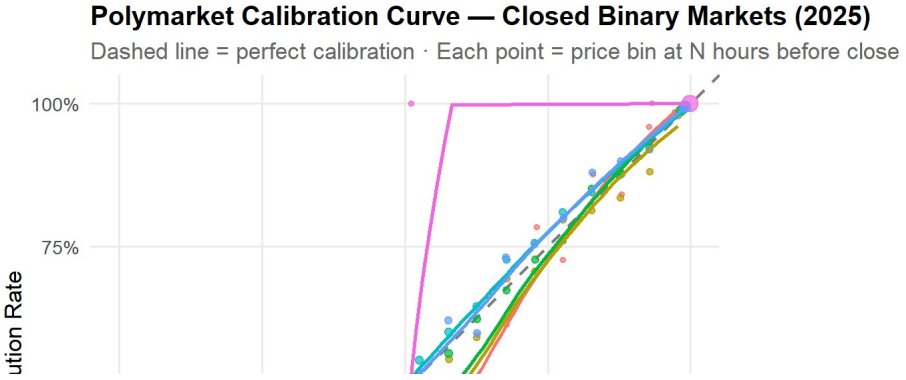
```
##
##  245827 total market × snapshot observations
```

Show

```
## 49753 markets with full price history stats
```

# 2.1 Overall Calibration Curve

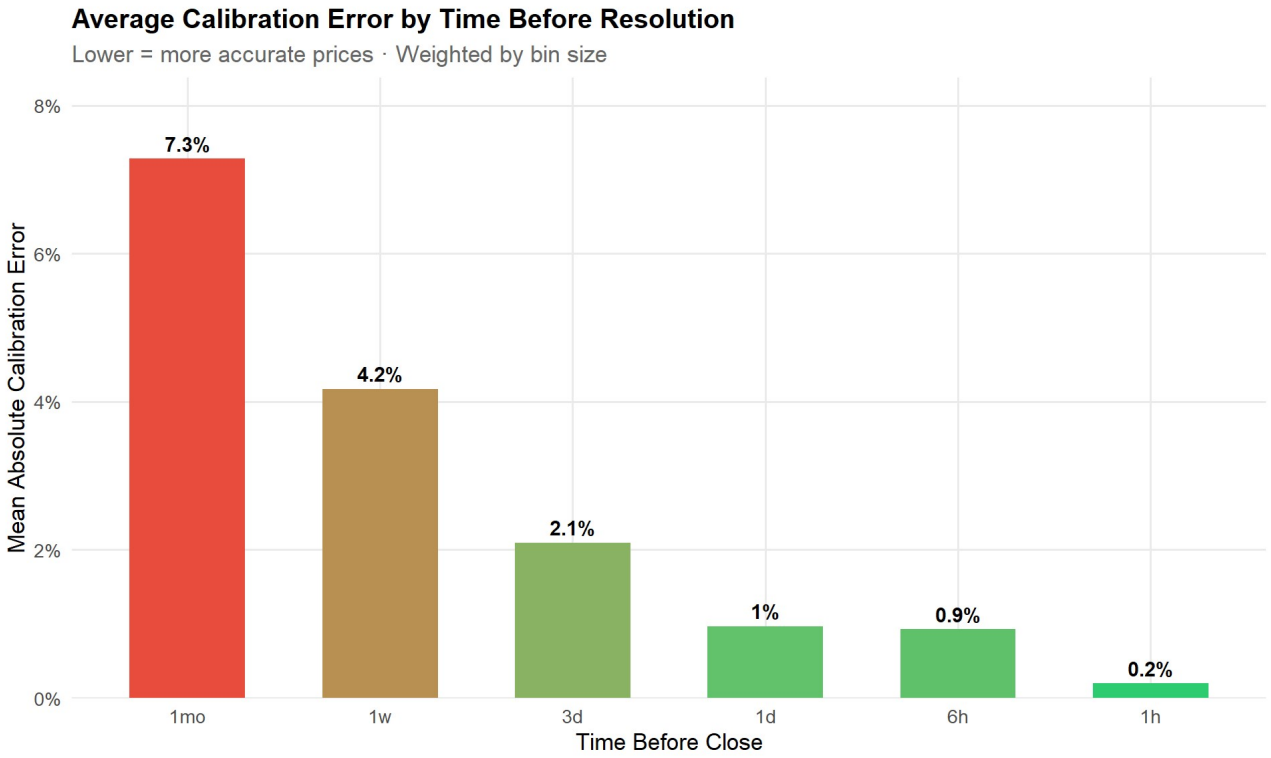
Show





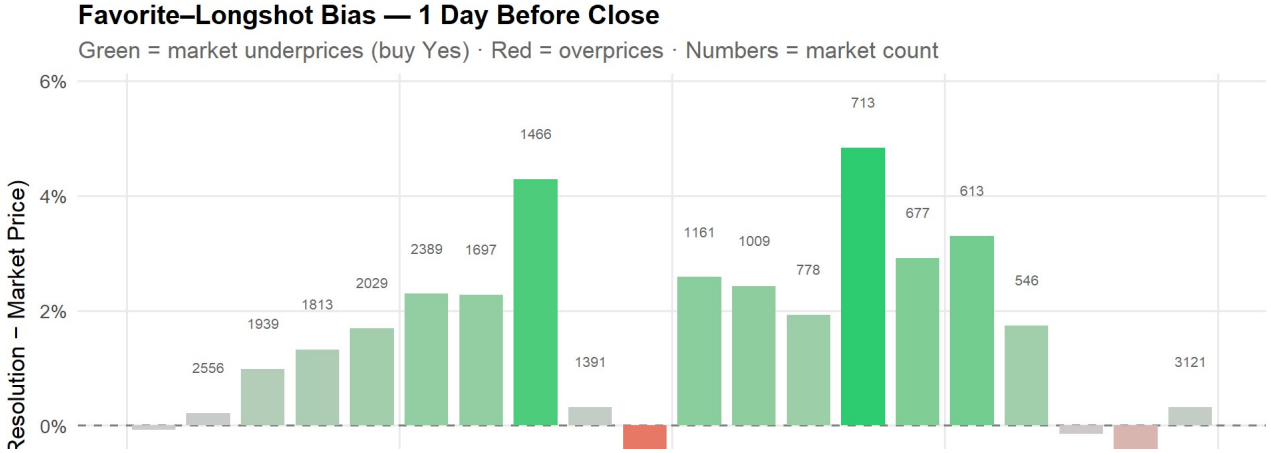

## 2.2 Calibration Error by Snapshot Horizon

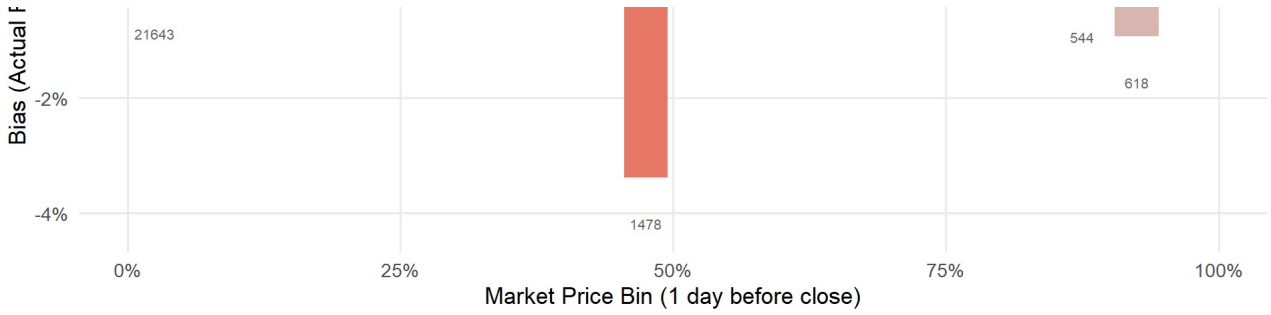
Show




## 2.3 Favorite–Longshot Bias

Show

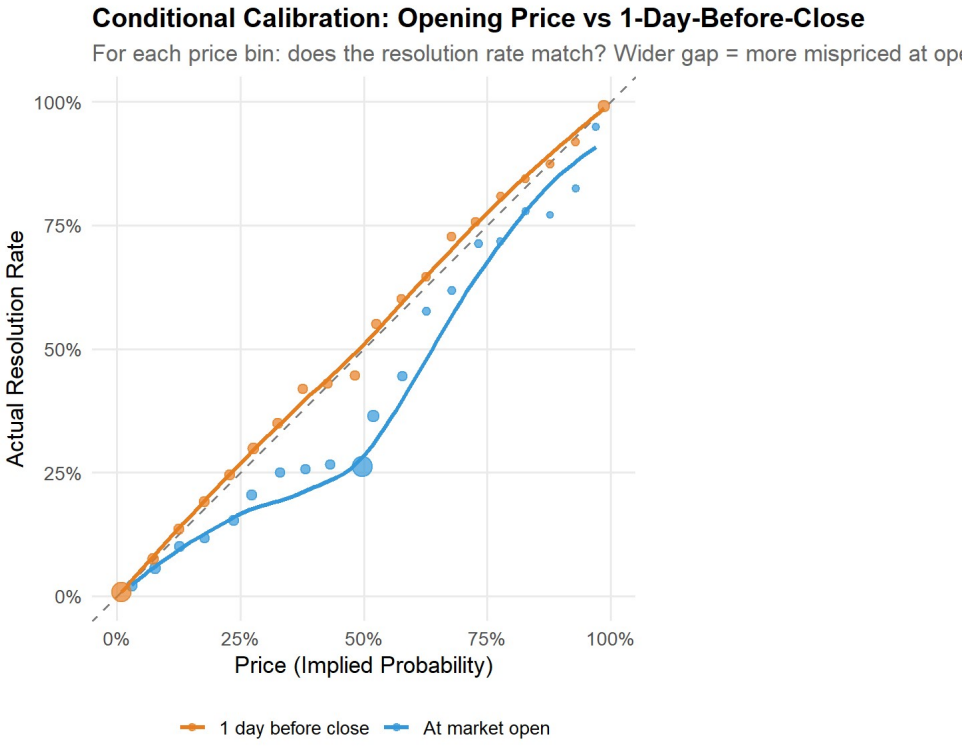




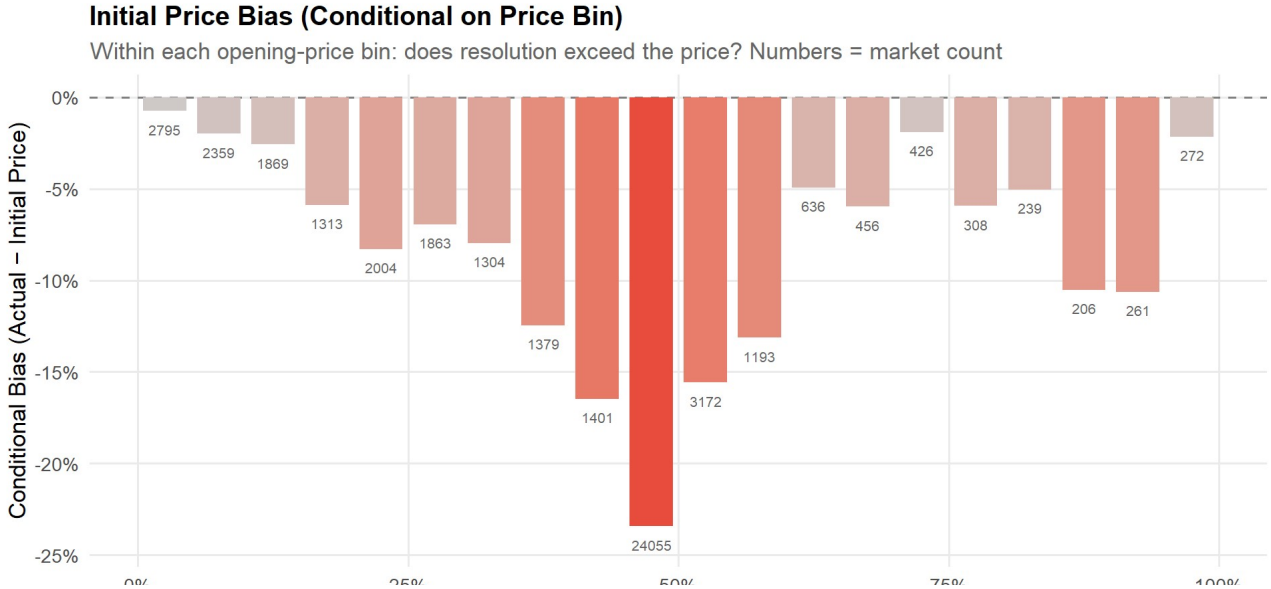

# 2.4 Initial Price Calibration (Conditional)

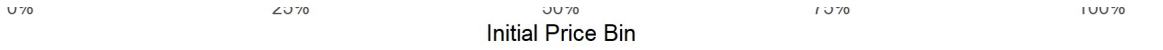
**Important**: The naive aggregate "initial bias" (initial price minus resolution averaged across all markets) is misleading because it conflates the base rate of Yes resolutions (≈27%) with a default opening price near 50%. The correct test is *conditional*: for markets **opening in a given price bin**, does the resolution rate match that bin? This isolates real initial mispricing from compositional effects.

Show




Show






Show

```
## ── Initial price calibration (conditional, within-bin) ──
```

Show

```
##   Weighted mean absolute bias (initial):    15.65%
```

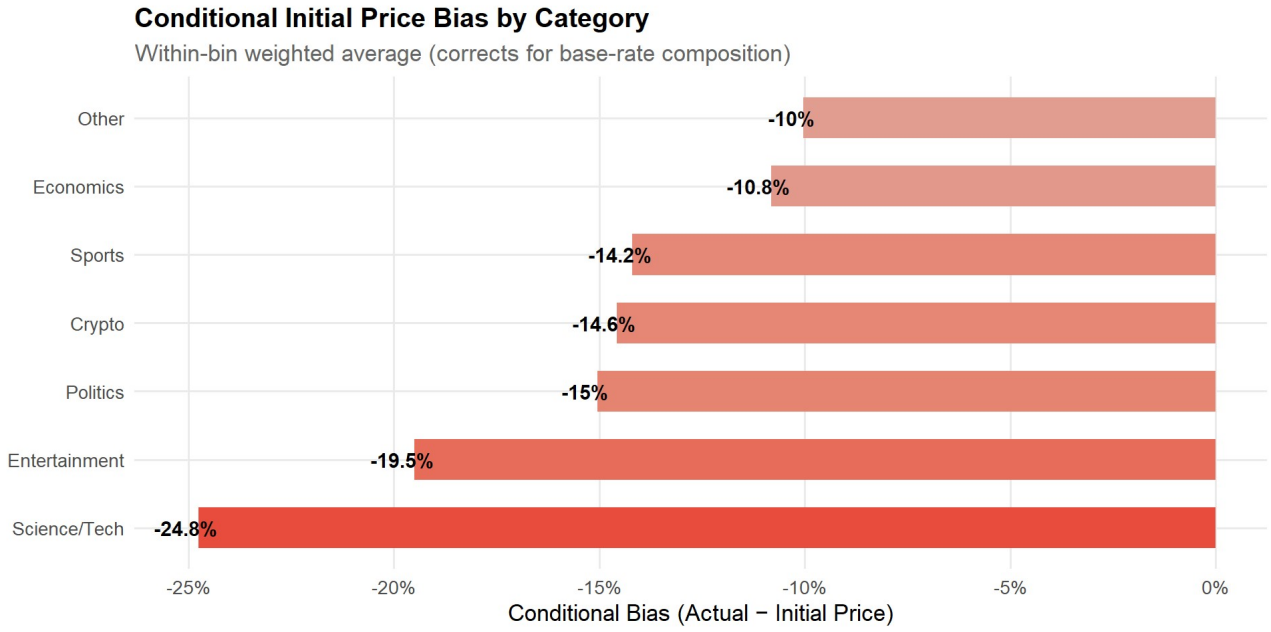
Show

```
##   Weighted mean absolute bias (1d close):   1.00%
```

Show

```
##   (If initial >> close, new markets are genuinely mispriced and correct over time)
```

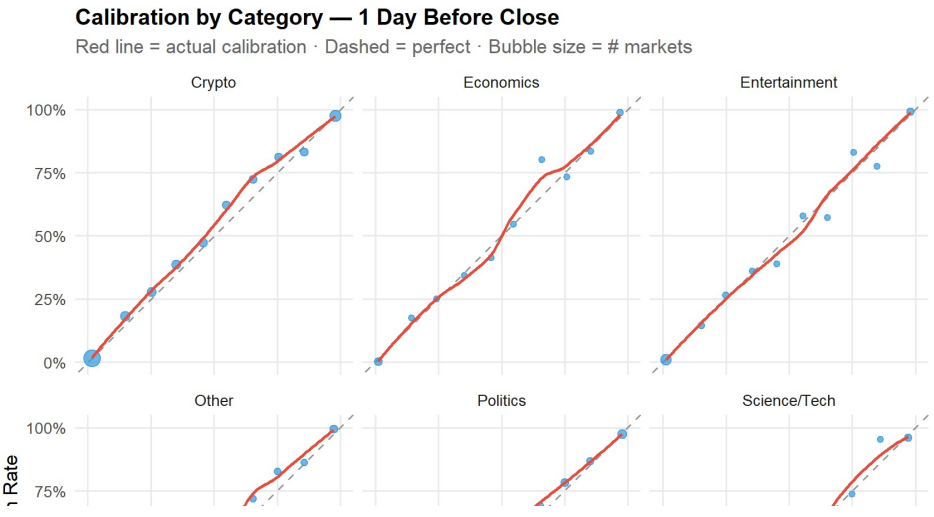
Show

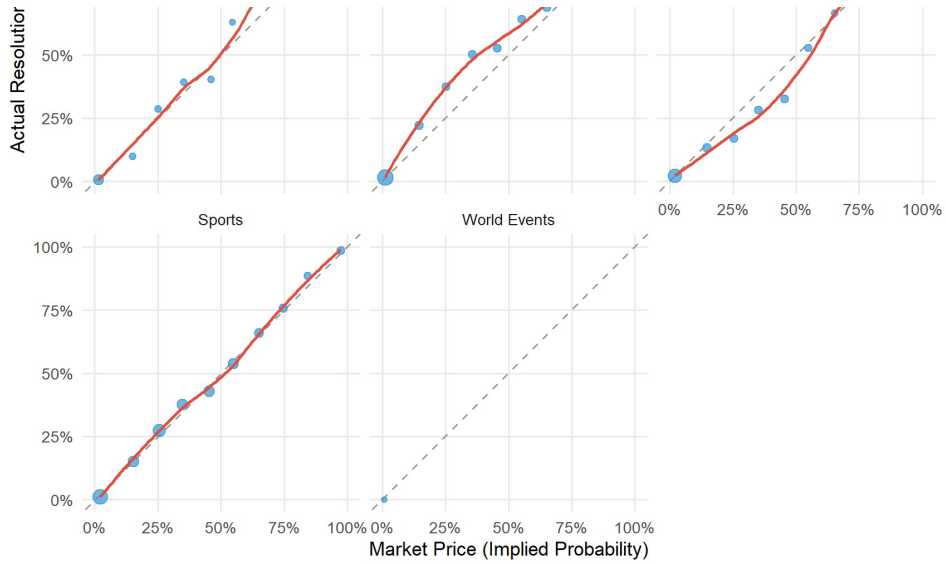



# 3 · Category-Level Mispricings

## 3.1 Category Calibration Curves

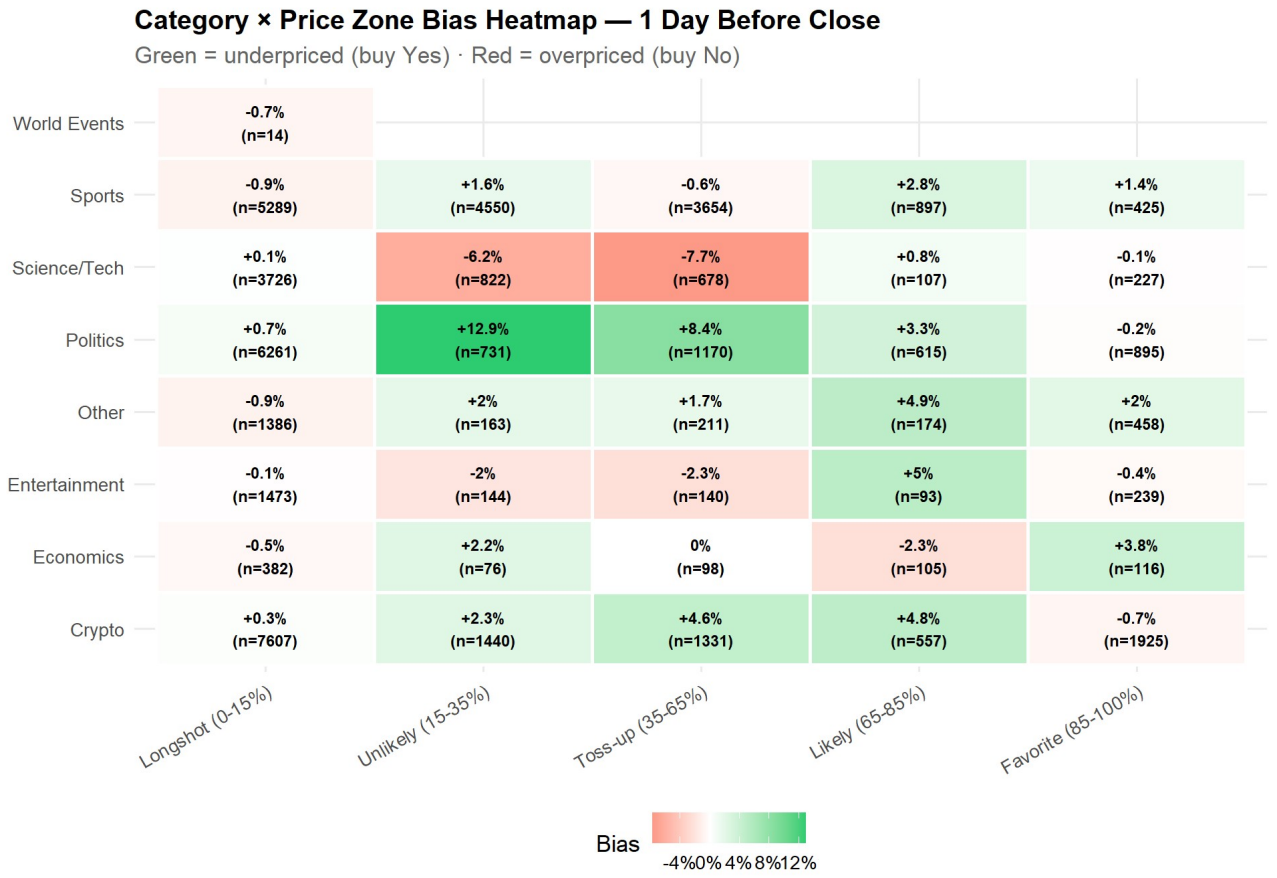
Show






# 3.2 Category × Price Zone Bias Heatmap

Show




# 3.3 Aggregate Category Bias & Brier Score

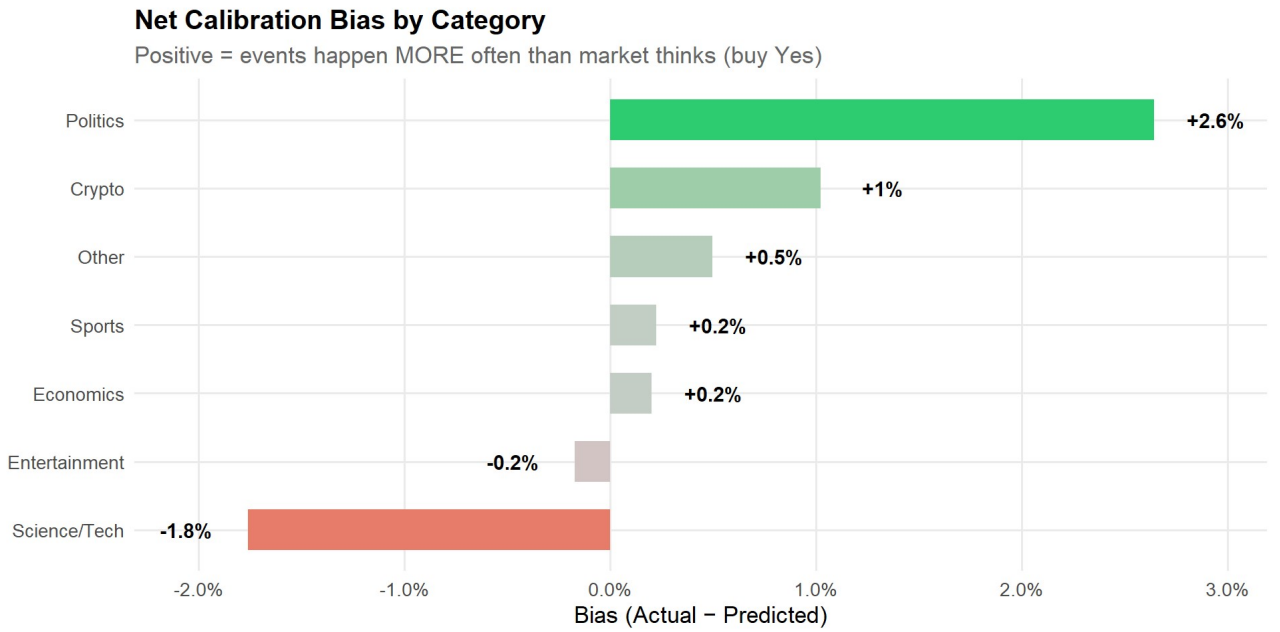
Show

Category-Level Bias & Accuracy (1 day before close)

| category | n_markets | avg_price | resolution_rate | bias | brier_score | abs_bias |
|---|---|---|---|---|---|---|
| World Events | 18 | 0.1137 | 0.2222 | 0.1085 | 0.0650 | 0.1193 |
| Politics | 9672 | 0.2265 | 0.2529 | 0.0264 | 0.0755 | 0.1431 |
| Science/Tech | 5560 | 0.1656 | 0.1480 | -0.0176 | 0.0743 | 0.1533 |

| category | n_markets | avg_price | resolution_rate | bias | brier_score | abs_bias |
|---|---|---|---|---|---|---|
| Crypto | 12860 | 0.2694 | 0.2796 | 0.0102 | 0.0732 | 0.1438 |
| Other | 2392 | 0.3098 | 0.3148 | 0.0050 | 0.0554 | 0.1200 |
| Sports | 14815 | 0.2823 | 0.2846 | 0.0022 | 0.1407 | 0.2826 |
| Economics | 777 | 0.3403 | 0.3423 | 0.0020 | 0.0846 | 0.1735 |
| Entertainment | 2089 | 0.2061 | 0.2044 | -0.0017 | 0.0473 | 0.0972 |

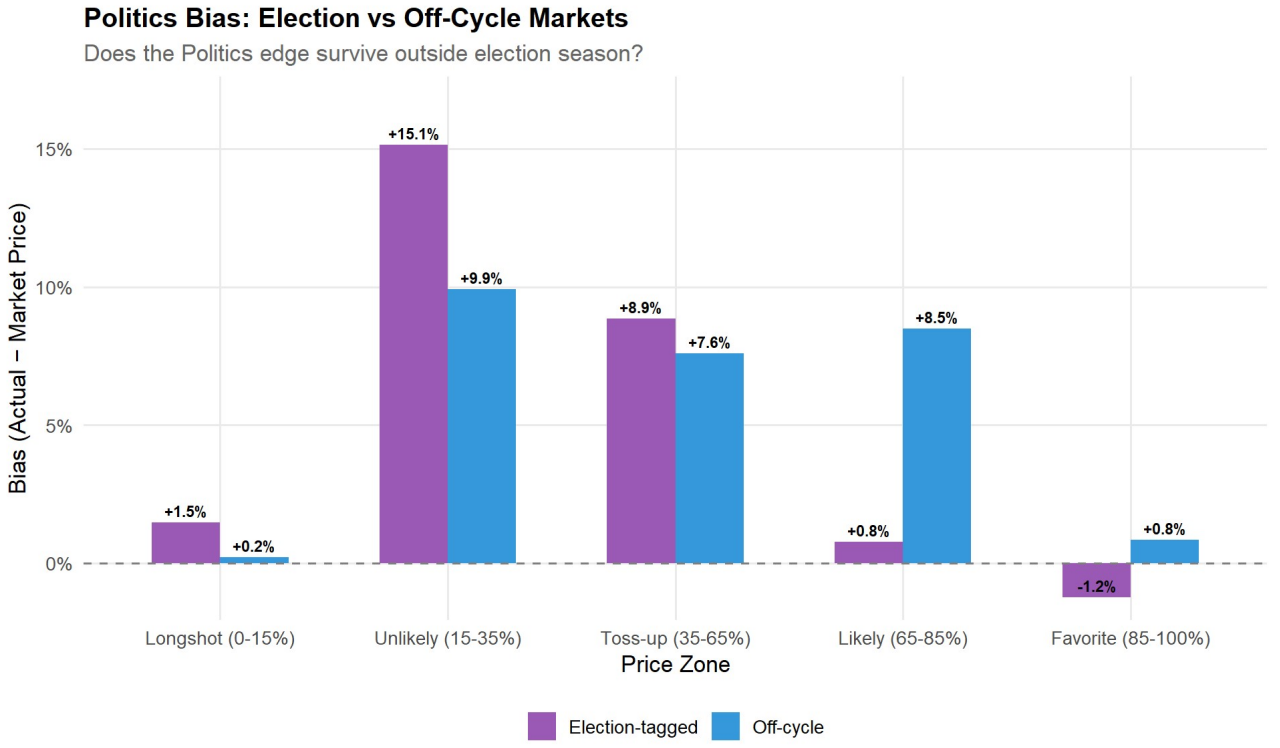
Show




# 3.4 Politics: Election vs Off-Cycle Markets

The Politics category tends to show the strongest aggregate bias. We check how much of this is 2024 US election spillover — if the edge is election-specific, it may not persist in off-cycle periods.

Show



Show

Politics: Election vs Off-Cycle Bias (1 day before close)

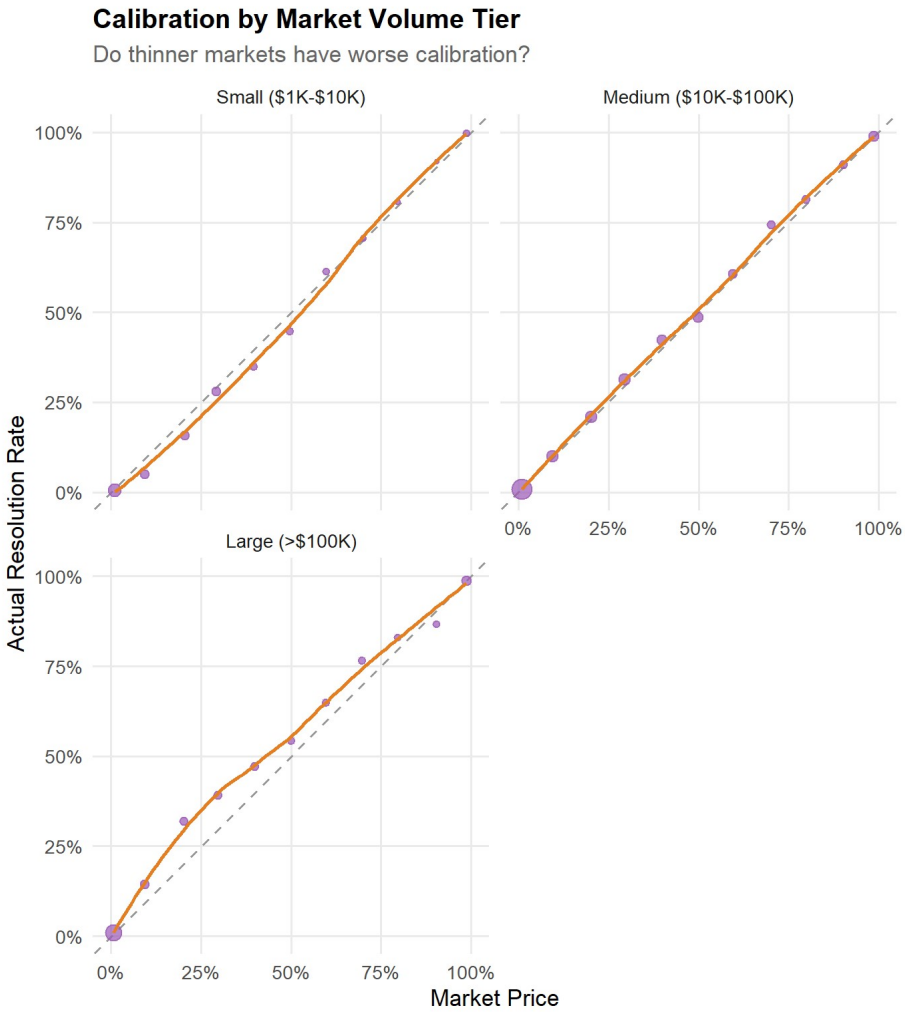
| market_type | n_markets | avg_price | resolution_rate | bias | brier_score |
|---|---|---|---|---|---|
| Election-tagged | 4476 | 0.2844 | 0.3213 | 0.0369 | 0.1054 |
| Off-cycle | 5196 | 0.1766 | 0.1940 | 0.0174 | 0.0497 |

# 4 · Volume & Liquidity Analysis

Thin markets are noisier. Let's check if bias patterns change when we weight by volume, and critically — which edges survive once we filter for markets a bot can actually trade.

## 4.1 Calibration by Volume Tier

Show




Show

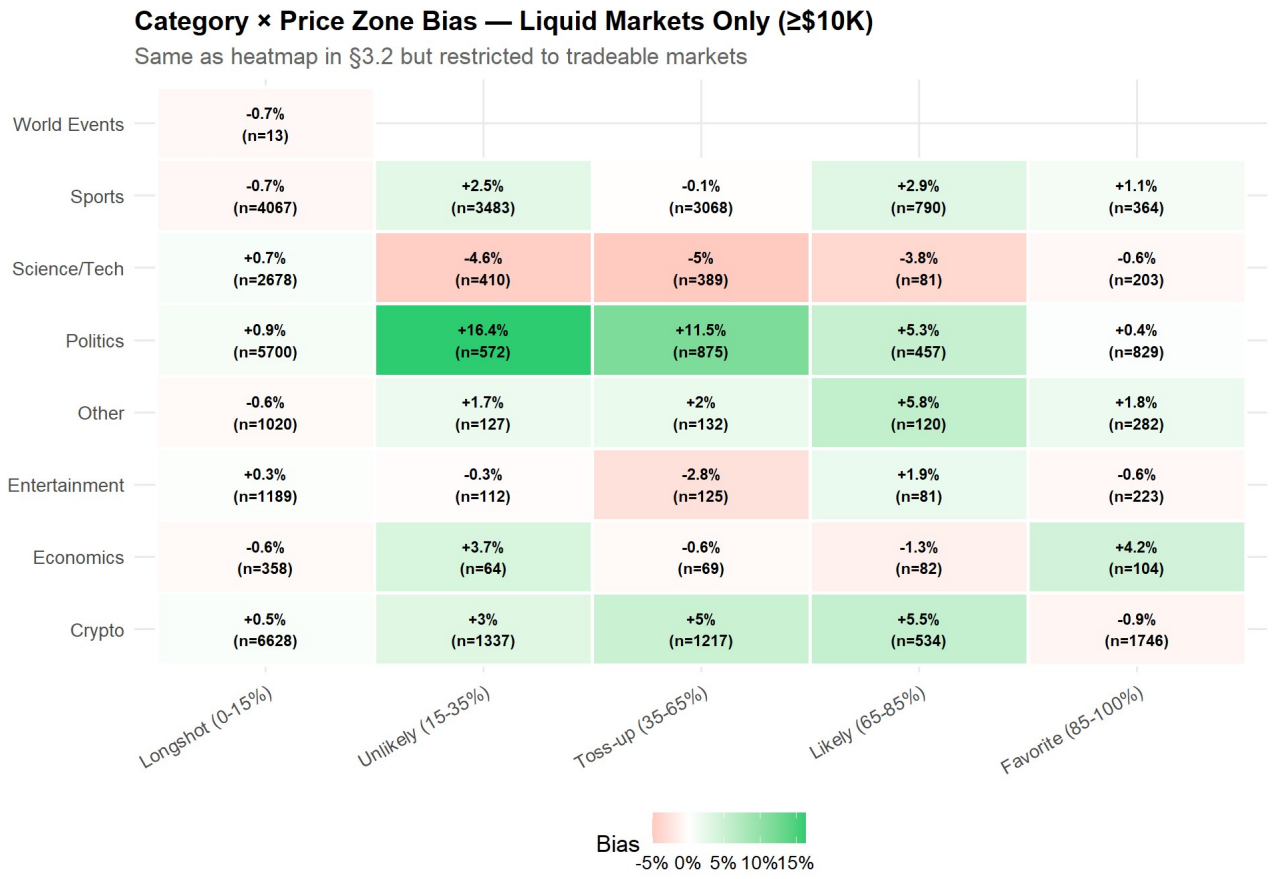
Brier Score & Bias by Volume Tier

| vol_tier | n | brier_score | bias |
|---|---|---|---|
| Small ($1K-$10K) | 8650 | 0.0917 | -0.0188 |
| Medium ($10K-$100K) | 26440 | 0.0958 | 0.0058 |
| Large (>$100K) | 13093 | 0.0872 | 0.0263 |

# 4.2 Liquidity-Filtered Edge Survival

**Critical for bot**: Edges in markets with < $10K volume are un-tradeable (slippage would eat the edge). Here we recompute the exploitable segment table restricted to liquid markets.

Show




Show

```
## ── Liquidity filter impact ──
```

Show

```
##   Exploitable segments (all markets):     10
```

Show

```
##   Exploitable segments (volume ≥ $10K):   11
```

Show

```
##   Survival rate:                          110%
```

# 5 · Market Duration Effects

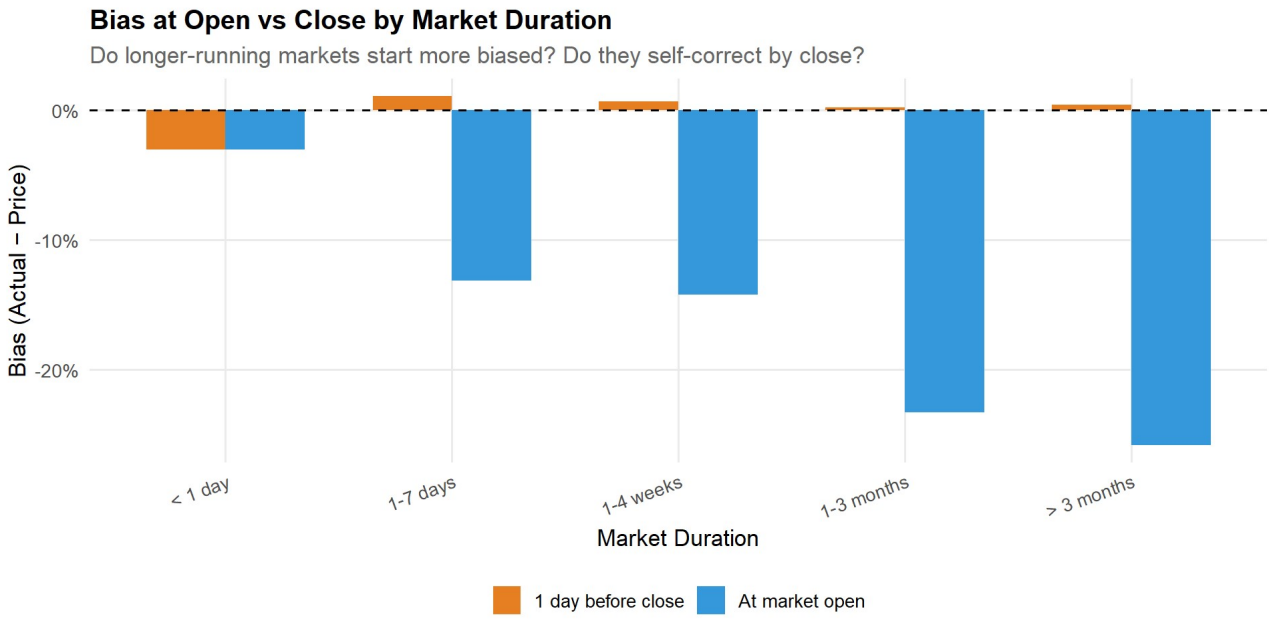
Show

Bias & Calibration by Market Duration

| duration_tier | n | init_bias | final_bias | price_range | resolution_r | brier_initial |
|---|---|---|---|---|---|---|
| < 1 day | 675 | -0.0300 | -0.0300 | 0.4651 | 0.4089 | 0.1925 |
| 1-7 days | 15075 | -0.1312 | 0.0109 | 0.5112 | 0.2825 | 0.2036 |

| duration_tier | n | init_bias | final_bias | price_range | resolution_r | brier_initial |
|---|---|---|---|---|---|---|
| 1-4 weeks | 24046 | -0.1422 | 0.0068 | 0.5304 | 0.2810 | 0.2057 |
| 1-3 months | 4915 | -0.2327 | 0.0025 | 0.4973 | 0.1727 | 0.1928 |
| > 3 months | 3472 | -0.2581 | 0.0044 | 0.4753 | 0.1129 | 0.1776 |

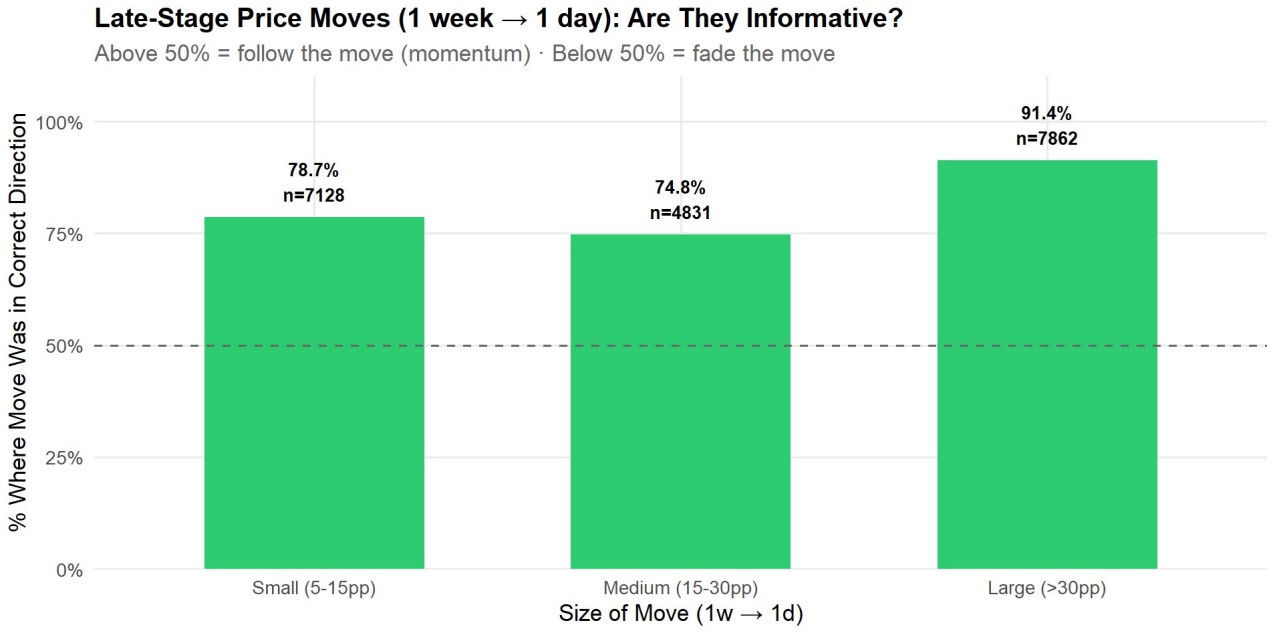
Show



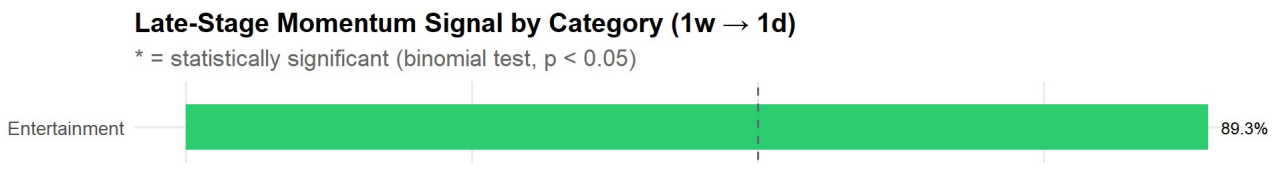

# 6 · Late-Stage Price Signals

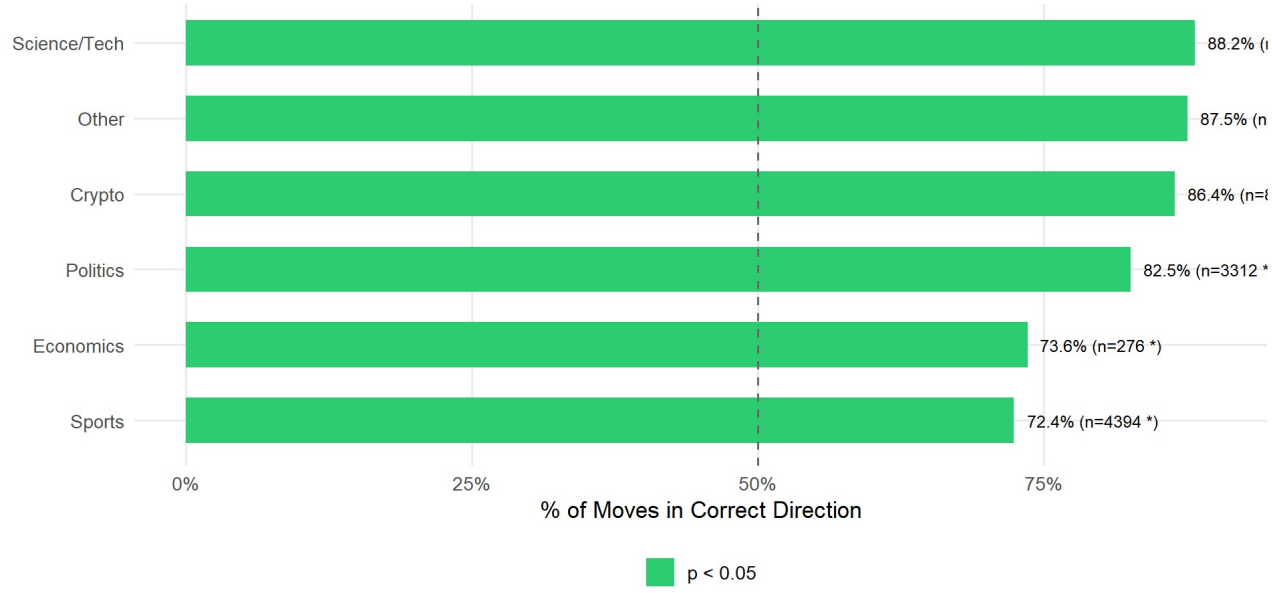
When the price moves substantially in the final week, is that move informative about the resolution? If yes, **following the move** (momentum) is a tradeable signal. If no (or reverse), **fading the move** is.

Show




Show





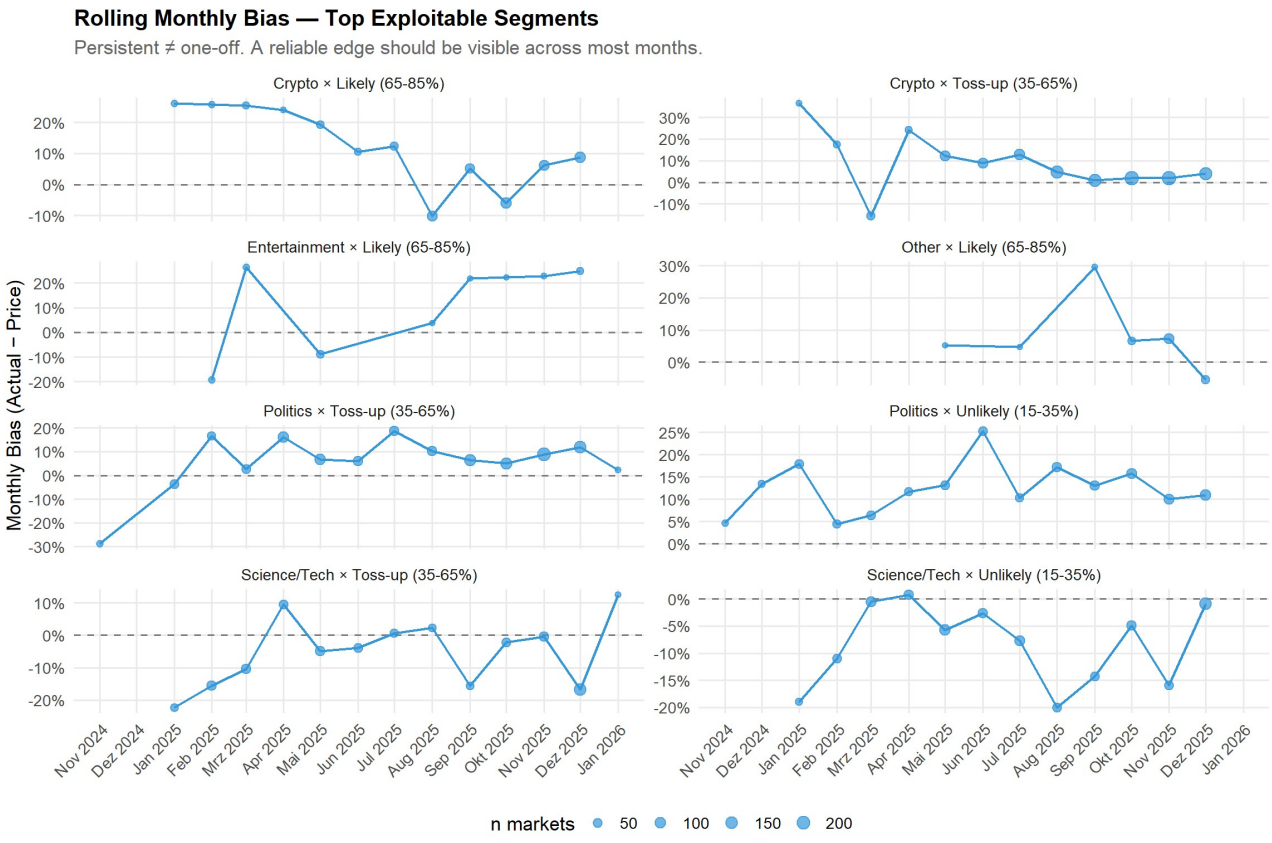

# 7 · Temporal Stability of Edges

**Critical for bot**: A bias that appeared once is noise; a bias that persists monthly is structural. Here we track the top exploitable segments over time to see which edges a bot can rely on.

Show

## 7.1 Rolling Monthly Bias for Top Segments

Show




## 7.2 Edge Persistence Score

Show

Edge Persistence: How Many Months Does the Bias Appear?

| segment | overall_bias | n_months | months_correct | persistence_pct | avg_monthly_n |
|---|---|---|---|---|---|
| Politics × Unlikely (15-35%) | 12.9% | 14 | 14 | 100% | 52 |
| Crypto × Toss-up (35-65%) | 4.6% | 12 | 11 | 92% | 111 |
| Science/Tech × Unlikely (15-35%) | -6.2% | 12 | 11 | 92% | 68 |
| Politics × Toss-up (35-65%) | 8.4% | 14 | 12 | 86% | 83 |
| Crypto × Likely (65-85%) | 4.8% | 12 | 10 | 83% | 46 |
| Other × Likely (65-85%) | 4.9% | 6 | 5 | 83% | 26 |
| Entertainment × Likely (65-85%) | 5% | 8 | 6 | 75% | 10 |
| Science/Tech × Toss-up (35-65%) | -7.7% | 13 | 9 | 69% | 52 |

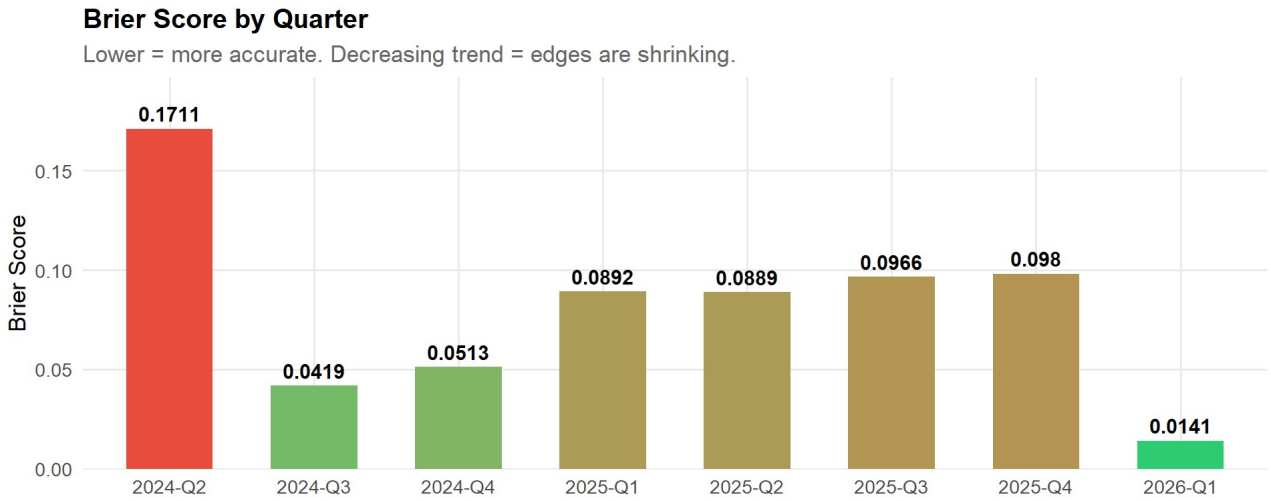
# 7.3 Is the Market Getting More Efficient?

Show

Calibration by Quarter — Is the market getting more efficient over time?

| close_quarter | n | bias | brier_score | abs_bias |
|---|---|---|---|---|
| 2024-Q2 | 6 | -0.0049 | 0.1711 | 0.2949 |
| 2024-Q3 | 9 | -0.0814 | 0.0419 | 0.0873 |
| 2024-Q4 | 350 | -0.0285 | 0.0513 | 0.1089 |
| 2025-Q1 | 5128 | 0.0008 | 0.0892 | 0.1770 |
| 2025-Q2 | 7207 | 0.0071 | 0.0889 | 0.1791 |
| 2025-Q3 | 10434 | 0.0079 | 0.0966 | 0.1923 |
| 2025-Q4 | 23703 | 0.0091 | 0.0980 | 0.1945 |
| 2026-Q1 | 1346 | -0.0060 | 0.0141 | 0.0309 |

Show




# 8 · Exploitable Pattern Summary

## 8.1 Gross Edge Table

Show

Top 20 Most Exploitable Category × Price Zone Combinations (|bias| > 3%, n ≥ 15)

| category | price_zone | n_markets | avg_price | actual_rate | bias | edge_pct | brier |
|---|---|---|---|---|---|---|---|
| Politics | Unlikely (15-35%) | 731 | 0.253 | 0.382 | 0.129 | 51.071 | 0.247 |
| Politics | Toss-up (35-65%) | 1170 | 0.502 | 0.586 | 0.084 | 16.802 | 0.247 |
| Science/Tech | Toss-up (35-65%) | 678 | 0.465 | 0.388 | -0.077 | -16.506 | 0.232 |
| Science/Tech | Unlikely (15-35%) | 822 | 0.251 | 0.189 | -0.062 | -24.751 | 0.155 |
| Entertainment | Likely (65-85%) | 93 | 0.756 | 0.806 | 0.050 | 6.656 | 0.152 |
| Other | Likely (65-85%) | 174 | 0.761 | 0.810 | 0.049 | 6.471 | 0.156 |
| Crypto | Likely (65-85%) | 557 | 0.751 | 0.799 | 0.048 | 6.371 | 0.164 |
| Crypto | Toss-up (35-65%) | 1331 | 0.482 | 0.527 | 0.046 | 9.444 | 0.243 |
| Economics | Favorite (85-100%) | 116 | 0.945 | 0.983 | 0.038 | 3.976 | 0.019 |
| Politics | Likely (65-85%) | 615 | 0.749 | 0.782 | 0.033 | 4.470 | 0.169 |

# 8.2 Transaction Cost–Adjusted Edge Table

**The critical filter**: gross edges are meaningless without accounting for the ~2% spread and ~2% winner fee. A segment is only tradeable if the net edge after costs exceeds ~1%.

Show

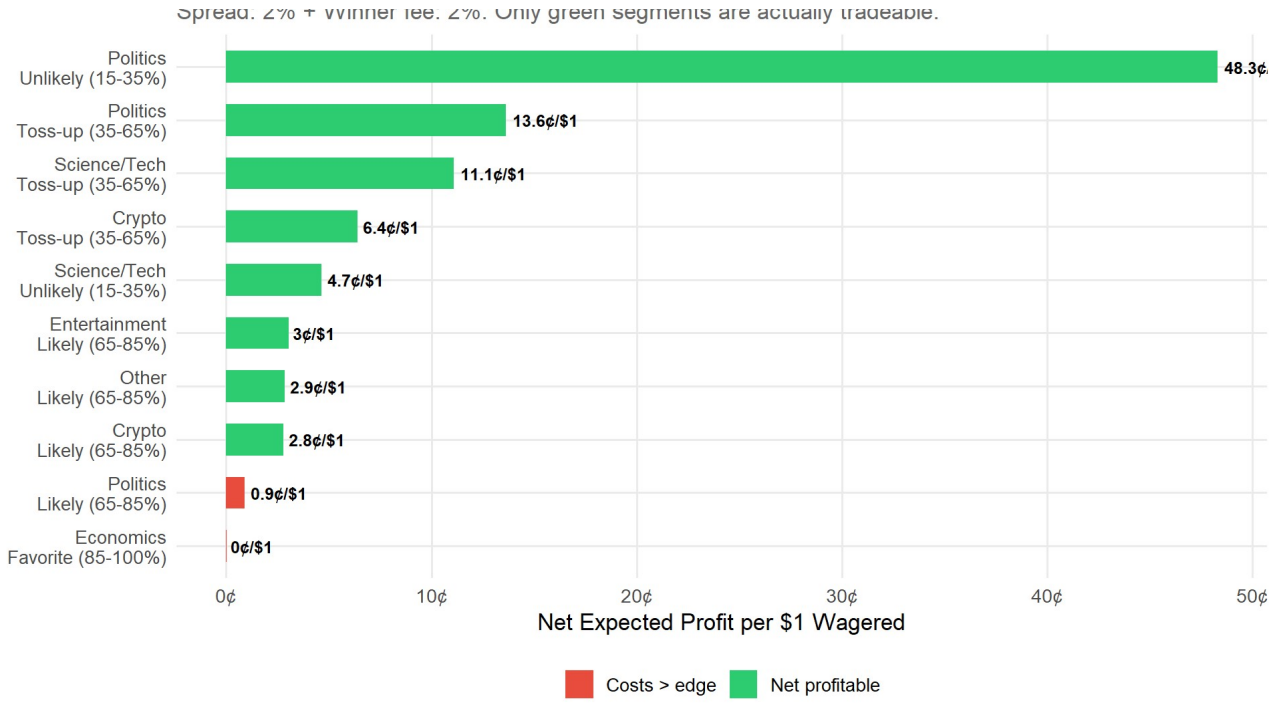
Cost-Adjusted Edges (spread=2%, winner fee=2%, min net edge=1%)

| category | price_zone | n_markets | strategy | gross_edge | spread | fee_cost | total_cost | net_edge | net_pl | tradeable |
|---|---|---|---|---|---|---|---|---|---|---|
| Politics | Unlikely (15-35%) | 731 | Buy Yes | 12.9% | 2% | 0.8% | 2.8% | 10.1% | 48.3% | ✅ YES |
| Politics | Toss-up (35-65%) | 1170 | Buy Yes | 8.4% | 2% | 1.2% | 3.2% | 5.3% | 13.6% | ✅ YES |
| Science/Tech | Toss-up (35-65%) | 678 | Buy No | 7.7% | 2% | 1.2% | 3.2% | 4.4% | 11.1% | ✅ YES |
| Science/Tech | Unlikely (15-35%) | 822 | Buy No | 6.2% | 2% | 1.6% | 3.6% | 2.6% | 4.7% | ✅ YES |
| Crypto | Toss-up (35-65%) | 1331 | Buy Yes | 4.6% | 2% | 1.1% | 3.1% | 1.5% | 6.4% | ✅ YES |
| Entertainment | Likely (65-85%) | 93 | Buy Yes | 5% | 2% | 1.6% | 3.6% | 1.4% | 3% | ✅ YES |
| Other | Likely (65-85%) | 174 | Buy Yes | 4.9% | 2% | 1.6% | 3.6% | 1.3% | 2.9% | ✅ YES |
| Crypto | Likely (65-85%) | 557 | Buy Yes | 4.8% | 2% | 1.6% | 3.6% | 1.2% | 2.8% | ✅ YES |
| Economics | Favorite (85-100%) | 116 | Buy Yes | 3.8% | 2% | 2% | 4% | -0.2% | 0% | ❌ NO |
| Politics | Likely (65-85%) | 615 | Buy Yes | 3.3% | 2% | 1.6% | 3.6% | -0.2% | 0.9% | ❌ NO |

Show

**Expected Net P&L per $1 Bet — After Transaction Costs**

Spread: 2% + Winner fee: 2% — Only green segments are actually tradeable




Show

```
## ── Cost filter impact ──
```

Show

```
##   Segments with gross |bias| > 3%:   10
```

Show

```
##   Segments profitable after costs:    8
```

Show

```
##   Kill rate:                          20%
```

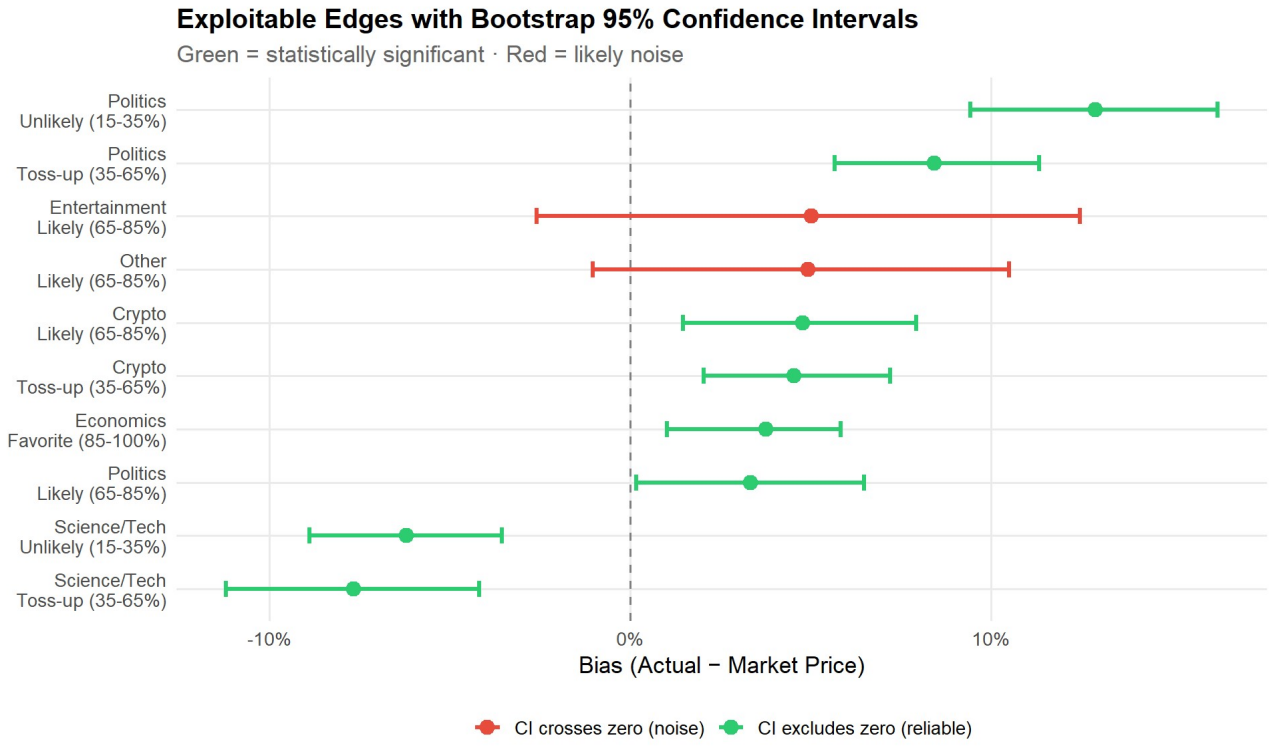
# 8.3 Bootstrap Confidence Intervals

Show

Bootstrap 95% CIs — Which Edges Are Statistically Real?

| category | price_zone | n_obs | point_bias | ci_lo | ci_hi | significant |
|---|---|---|---|---|---|---|
| Politics | Unlikely (15-35%) | 731 | 12.9% | 9.44% | 16.29% | ✅ YES |
| Politics | Toss-up (35-65%) | 1170 | 8.43% | 5.68% | 11.34% | ✅ YES |
| Science/Tech | Toss-up (35-65%) | 678 | -7.67% | -11.19% | -4.18% | ✅ YES |
| Science/Tech | Unlikely (15-35%) | 822 | -6.2% | -8.88% | -3.55% | ✅ YES |
| Crypto | Toss-up (35-65%) | 1331 | 4.55% | 2.05% | 7.21% | ✅ YES |
| Entertainment | Likely (65-85%) | 93 | 5.03% | -2.58% | 12.48% | no |
| Other | Likely (65-85%) | 174 | 4.93% | -1.04% | 10.5% | no |
| Crypto | Likely (65-85%) | 557 | 4.79% | 1.46% | 7.93% | ✅ YES |
| Economics | Favorite (85-100%) | 116 | 3.76% | 1.02% | 5.84% | ✅ YES |
| Politics | Likely (65-85%) | 615 | 3.35% | 0.18% | 6.49% | ✅ YES |

Show




# 8.4 Kelly Criterion Position Sizing

For each tradeable edge, optimal bet sizing via the Kelly criterion. Half-Kelly is recommended in practice (accounts for edge estimation error).

Show

Kelly Criterion Sizing — Tradeable Segments Only (after costs)

| category | price_zone | strategy | n_markets | net_edge | win_prob | kelly_full | kelly_half | growth_rate |
|---|---|---|---|---|---|---|---|---|
| Politics | Unlikely (15-35%) | Buy Yes | 731 | 10.1% | 38.2% | 16.5% | 8.2% | 2.729% |
| Politics | Toss-up (35-65%) | Buy Yes | 1170 | 5.3% | 58.6% | 14.1% | 7% | 0.695% |
| Science/Tech | Toss-up (35-65%) | Buy No | 678 | 4.4% | 61.2% | 13.2% | 6.6% | 0.529% |
| Science/Tech | Unlikely (15-35%) | Buy No | 822 | 2.6% | 81.1% | 15.2% | 7.6% | 0.255% |
| Crypto | Toss-up (35-65%) | Buy Yes | 1331 | 1.5% | 52.7% | 6% | 3% | 0.138% |
| Entertainment | Likely (65-85%) | Buy Yes | 93 | 1.4% | 80.6% | 10.2% | 5.1% | 0.109% |
| Other | Likely (65-85%) | Buy Yes | 174 | 1.3% | 81% | 9.8% | 4.9% | 0.098% |
| Crypto | Likely (65-85%) | Buy Yes | 557 | 1.2% | 79.9% | 9% | 4.5% | 0.087% |

# 8.5 Composite Exploitability Score

A single score the bot can compute in real-time for any market. Combines calibration bias, liquidity, statistical significance, and temporal persistence.

Show

Composite Exploitability Score (0-1) — Higher = more reliably tradeable

| segment | strategy | net_edge | edge_score | stat_score | persist_score | sample_score | composite_score |
|---|---|---|---|---|---|---|---|
| Politics × Unlikely (15-35%) | Buy Yes | 10.1% | 1.00 | 1 | 1.00 | 1.00 | 1.00 |

| segment | strategy | net_edge | edge_score | stat_score | persist_score | sample_score | composite_score |
|---|---|---|---|---|---|---|---|
| Politics × Toss-up (35-65%) | Buy Yes | 5.3% | 0.53 | 1 | 0.86 | 1.00 | 0.80 |
| Science/Tech × Toss-up (35-65%) | Buy No | 4.4% | 0.44 | 1 | 0.69 | 1.00 | 0.73 |
| Science/Tech × Unlikely (15-35%) | Buy No | 2.6% | 0.26 | 1 | 0.92 | 1.00 | 0.72 |
| Crypto × Toss-up (35-65%) | Buy Yes | 1.5% | 0.15 | 1 | 0.92 | 1.00 | 0.68 |
| Crypto × Likely (65-85%) | Buy Yes | 1.2% | 0.12 | 1 | 0.83 | 1.00 | 0.65 |
| Other × Likely (65-85%) | Buy Yes | 1.3% | 0.13 | 0 | 0.83 | 0.35 | 0.31 |
| Entertainment × Likely (65-85%) | Buy Yes | 1.4% | 0.14 | 0 | 0.75 | 0.19 | 0.27 |

# 9 · Simulated Backtest

**The ultimate test**: if we had traded every qualifying market in the exploitable segments, what would the P&L look like?

Show

```
## ── Backtest universe ──
```

Show

```
##   Tradeable segments:  8
```

Show

```
##   Total qualifying bets: 5556
```

Show

```
##   Win rate:              61.4%
```

Show

```
##   Gross P&L (total):     $396.46 per $1/bet
```

Show

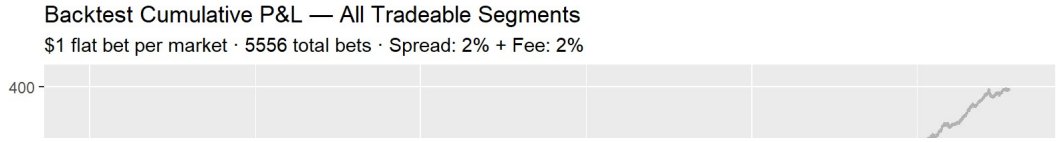
```
##   Net P&L (total):       $217.14 per $1/bet
```

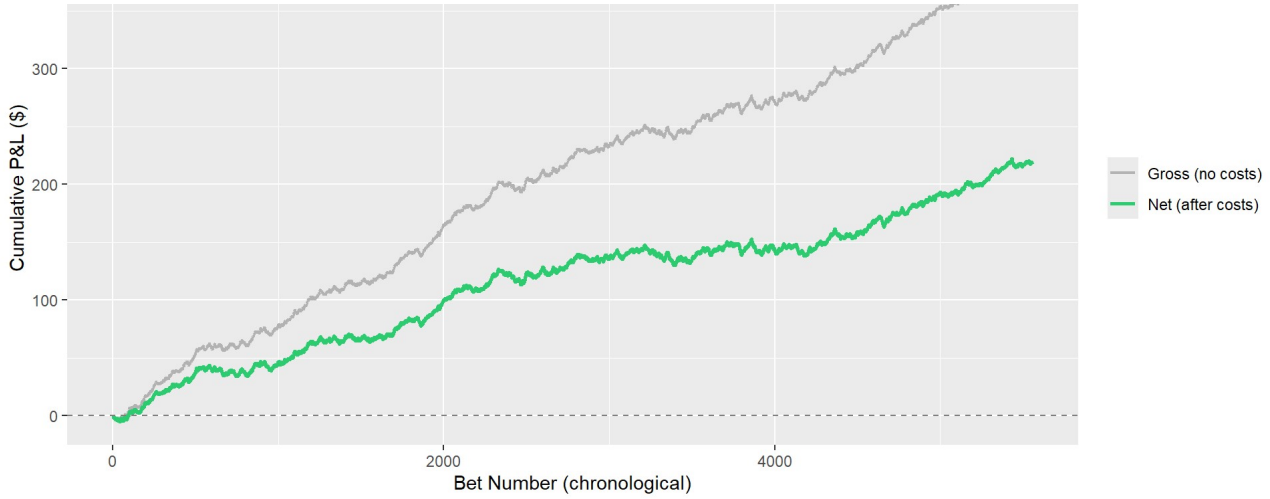
Show

```
##   Net P&L per bet:       3.91¢
```

# 9.1 Cumulative P&L

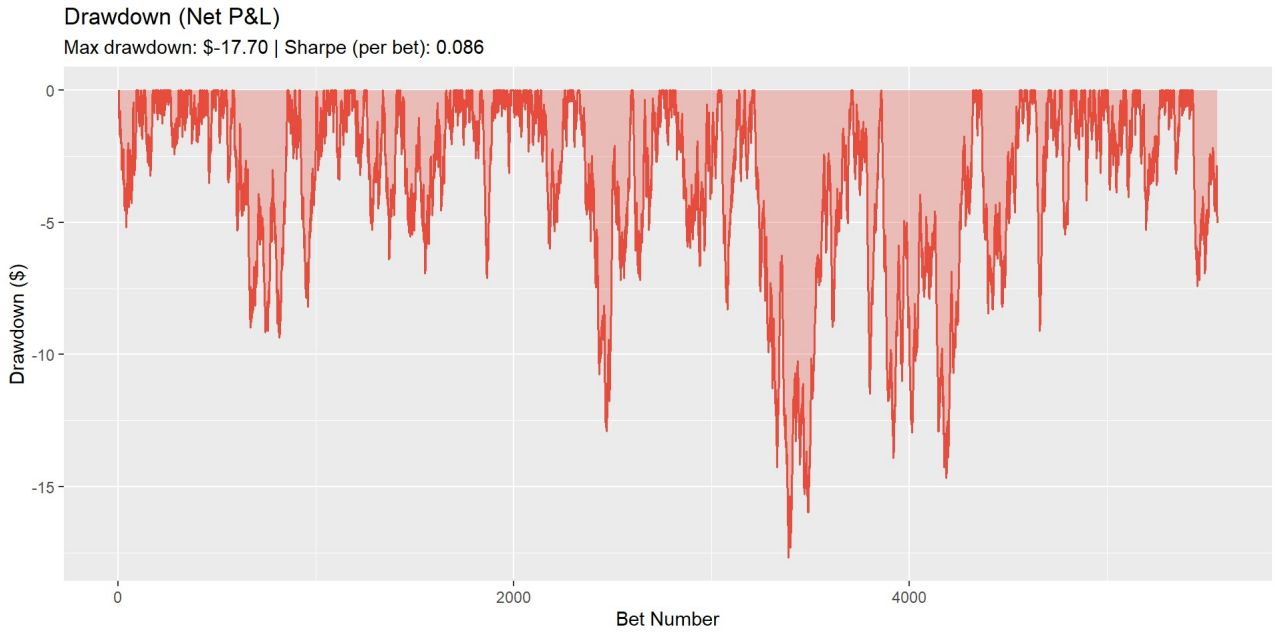
Show






# 9.2 Drawdown & Risk Metrics

Show

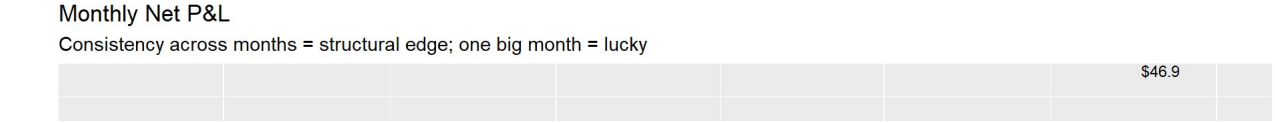



Show

Backtest Summary — Flat $1 Bets on All Tradeable Segments

| Metric | Value |
|---|---|
| Total bets | 5556 |
| Win rate | 61.4% |
| Gross P&L | $396.46 |
| Net P&L | 217.14 \| \|Net P&L per bet \|3.91¢ \| \|Std dev per bet \|45.22¢ \| \|Sharpe (per bet) \|0.086 \| \|Max drawdown \| -17.7 |
| Profitable months | 13/16 |

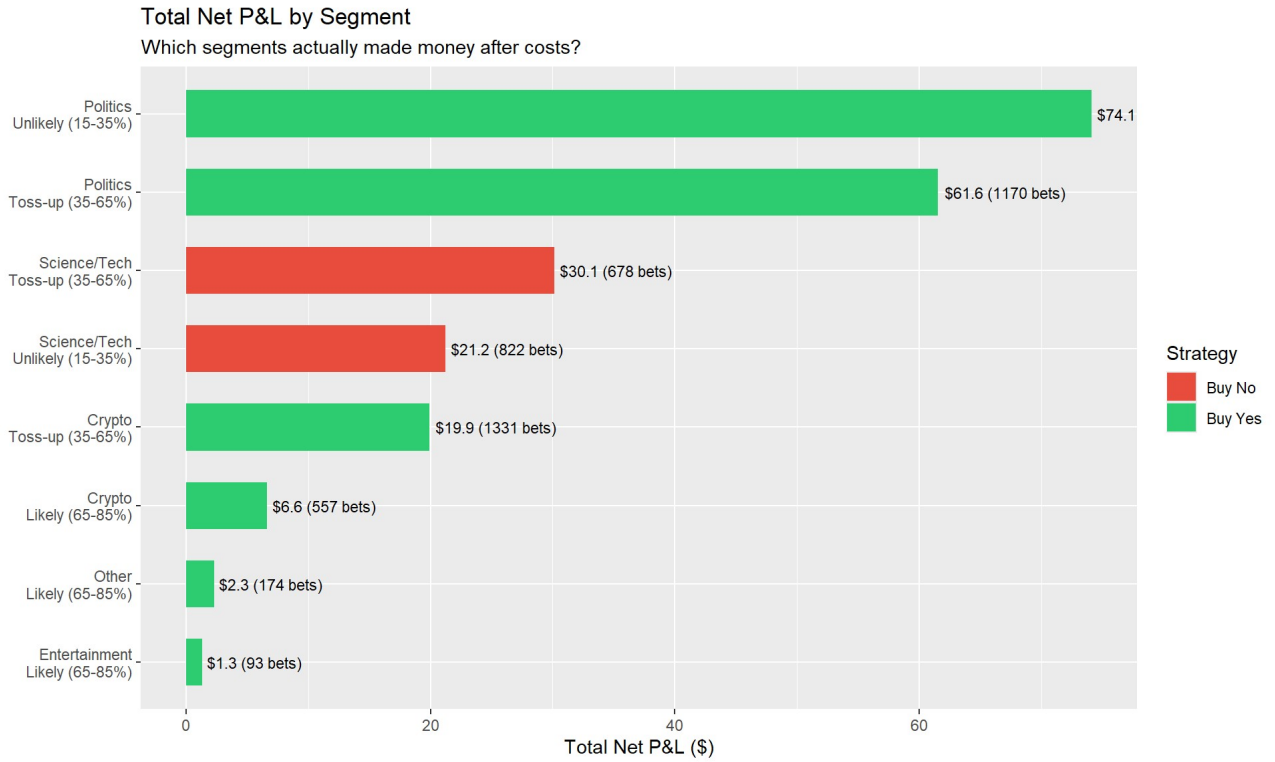
Show






# 9.3 P&L by Segment

Show




# 10 · Bot Implementation Notes

Show

## Entry Signals

Show

The bot should evaluate each new binary market against these criteria:

Show

1. **Category**: Match market tags to category rules

Show

2. **Price zone**: Current Yes price maps to one of 5 zones

Show

3. **Volume threshold**: Only trade markets with ≥ $10K volume

Show

4. **Net edge threshold**: Only enter if estimated net edge > 1%

Show

5. **Composite score**: Prefer segments with composite score > 0.5

Show

# Position Sizing

Show

- Use **half-Kelly** fractions from §8.4

Show

- Cap any single position at 5% of bankroll

Show

- Cap total exposure at 30% of bankroll (diversification)

Show

# Risk Limits

Show

- **Daily loss limit**: Pause trading after 5% bankroll drawdown in a day

Show

- **Monthly review**: Recompute all bias tables monthly with fresh data

Show

- **Edge decay**: If a segment's rolling 30-day bias flips sign, stop trading it

Show

- **Minimum sample**: Don't trade a segment with < 30 historical resolved markets

Show

# Key Findings

Show

- **Overall market bias** (1d before close): +0.7%. Markets are approximately well-calibrated overall.

Show

- **Tradeable edges after costs**: 8 segments survive (out of 10 gross)

Show

- **Statistically significant edges**: 8/10 top segments have bootstrap CIs excluding zero

Show

- **Best net edge**: Politics × Unlikely (15-35%) (Buy Yes, net edge +10.1%, n=731)

Show

- **Backtest result**: $217.14 net P&L on 5556 $1 bets (3.91¢ per bet avg)

Show

- **Max drawdown**: $-17.70 · Sharpe per bet: 0.086

Show

- **Most persistent edge**: Politics × Unlikely (15-35%) — bias in correct direction 100% of months

Show

Show

> **Disclaimer**: Past calibration patterns do not guarantee future exploitability.

Show

Transaction costs, slippage, and market evolution may erode these edges.

Show

Always paper-trade or use minimal size before scaling.

Show

The cost assumptions above ( 2 % spread, 2 % winner fee) are estimates — actual costs depend on market conditions and order size.

---

*Analysis generated on 2026-02-19 using 49753 closed binary markets and price history data.*

*v4 changes: fixed initial price bias (conditional analysis), added transaction cost model, temporal stability testing, liquidity filtering, Kelly sizing, composite scoring, simulated backtest. Removed: breakout analysis (null signal), DOW×category (multiple comparisons noise), monthly resolution rates, asymmetric error, price drift overview.*